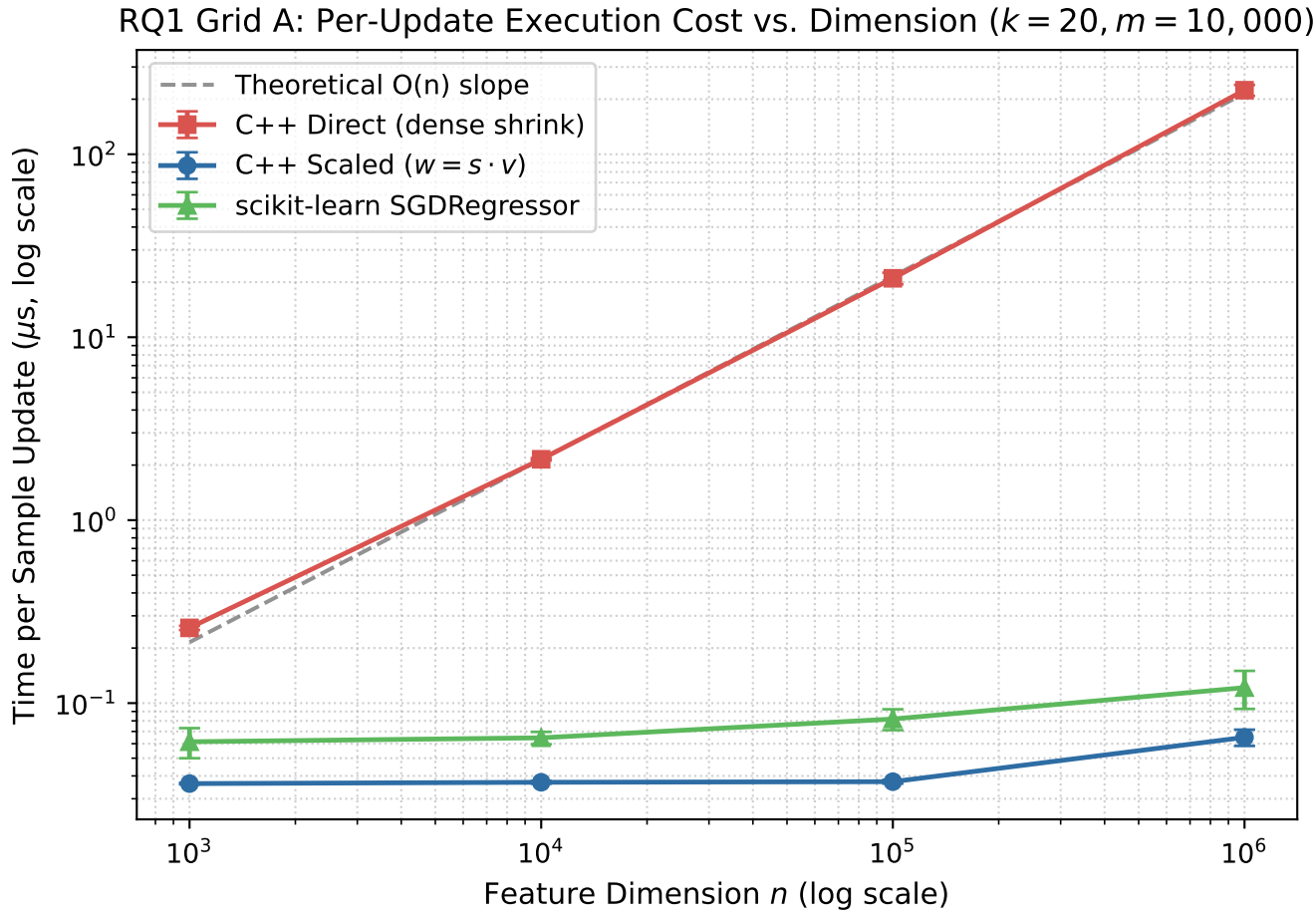
RQ1 Grid A: Per-Update Execution Cost vs. Dimension ($k = 20, m = 10,000$)
Theoretical O(n) slope
C++ Direct (dense shrink)
C++ Scaled ($w = s \cdot v$)
scikit-learn SGDRegressor
Time per Sample Update (μs, log scale)
10^{2}
10^{1}
10^{0}
10^{-1}
10^{3}
10^{4}
10^{5}
10^{6}
Feature Dimension n (log scale)